AN INTRODUCTION TO COLORED COINS

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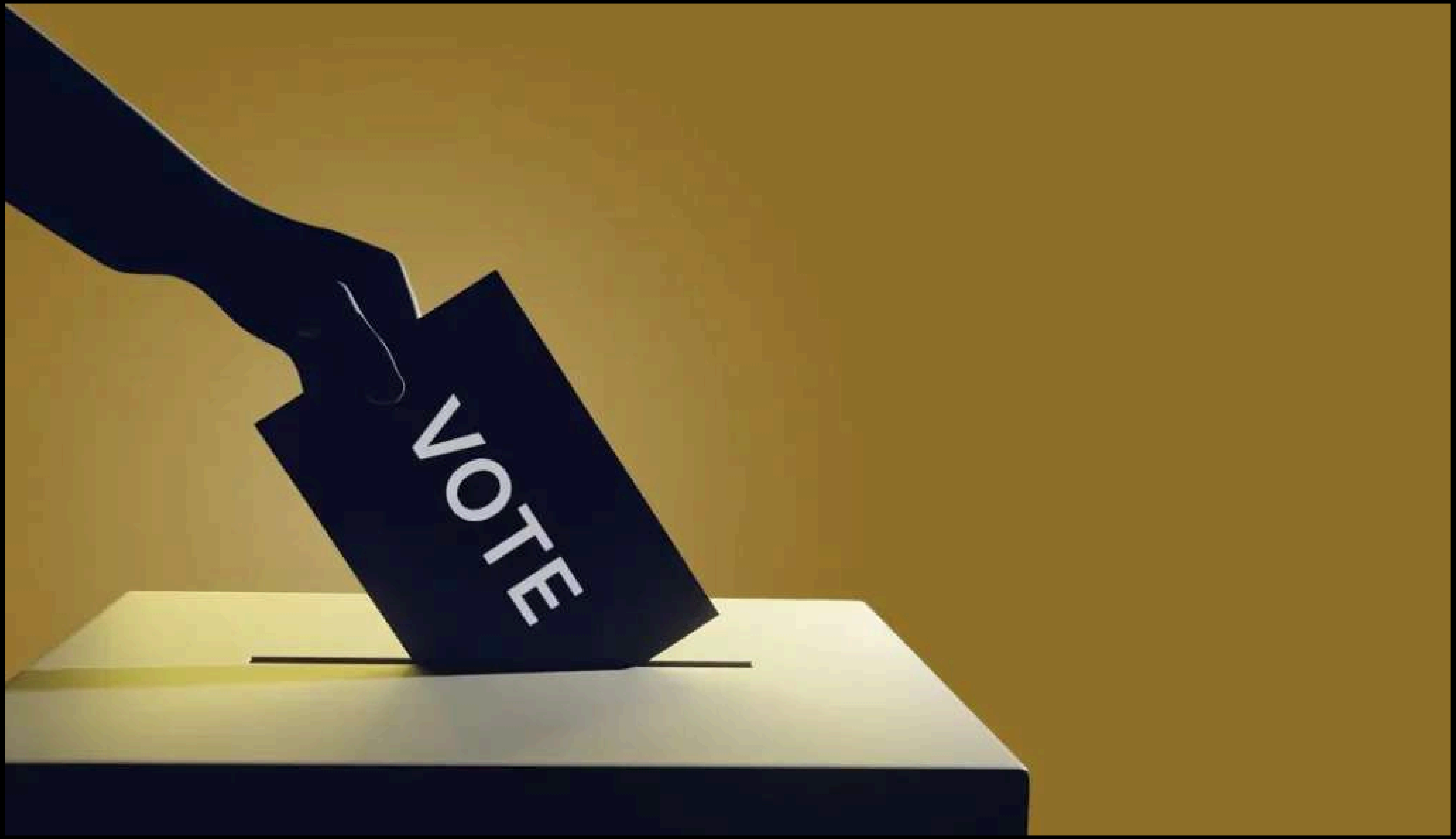
USING BLOCKCHAIN AS INFRASTRUCTURE

- Safety
- Decentralization
- Anonymity
- Availability

COLORED COINS

ISSUING SHARES

VOTING



SMART PROPERTY

COMMODITIES

OTHER USECASES

- Coupons
- Digital Collectibles
- Intellectual Property Rights
- Access/Subscription


EPOBC PROTOCOL
ethereum
ripple

EPOBC

- The first Colored coin implementation
- Zero memory overhead
- Distinguished from other bitcoin transactions by Sequence field
- 0x33 is the tag of a transfer transaction
- 0x25 is the tag of a genesis transaction

BITCOIN TRANSACTION FORMAT

Transaction

Field	Example	Size	Format	Description
Version ↘	`02000000`	4 bytes	Little-Endian	The version number for the transaction. Used to enable new features.
Marker ↘	`00`	1 byte		Used to indicate a segwit transaction. Must be `00`.
Flag ↘	`01`	1 byte		Used to indicate a segwit transaction. Must be `01` or greater.
Input Count ↘	`01`	*variable*	Compact Size	Indicates the number of inputs.
Input(s) ↘				
Output Count ↘	`02`	*variable*	Compact Size	Indicates the number of outputs.
Output(s) ↘				
Witness ↘				
Locktime ↘	`00000000`	4 bytes	Little-Endian	Set a time or height after which the transaction can be mined.

Input(s) ↘

Field	Example	Size	Format	Description
TXID ↘	`[TXID]`	32 bytes	Natural Byte Order	The TXID of the transaction containing the output you want to spend.
VOUT ↘	`01000000`	4 bytes	Little-Endian	The index number of the output you want to spend.
ScriptSig Size ↘	`6b`	*variable*	Compact Size	The size in bytes of the upcoming ScriptSig.
ScriptSig ↘	`[script]`	*variable*	Script	The unlocking code for the output you want to spend.
Sequence ↘	`fdffffff`	4 bytes	Little-Endian	Set whether the transaction can be replaced or when it can be mined.

This structure repeats for every input.

Output(s) ↘

Field	Example	Size	Format	Description
Amount ↘	`e99e060000000000`	8 bytes	Little-Endian	The value of the output in satoshis.
ScriptPubKey Size ↘	`19`	*variable*	Compact Size	The size in bytes of the upcoming ScriptPubKey.
ScriptPubKey ↘	`[script]`	*variable*	Script	The locking code for this output.

This structure repeats for every output.

Witness ↘

Field	Example	Size	Format	Description
Stack Items	`02`	*variable*	Compact Size	The number of items to be pushed on to the stack as part of the unlocking code.

Field	Example	Size	Format	Description
Size	`47`	*variable*	Compact Size	The size of the upcoming stack item.
Item	`304...b01`	*variable*	Bytes	The data to be pushed on to the stack.

This structure repeats for every stack item.

This structure repeats for every input.

SEQUENCE NUMBER FIELD

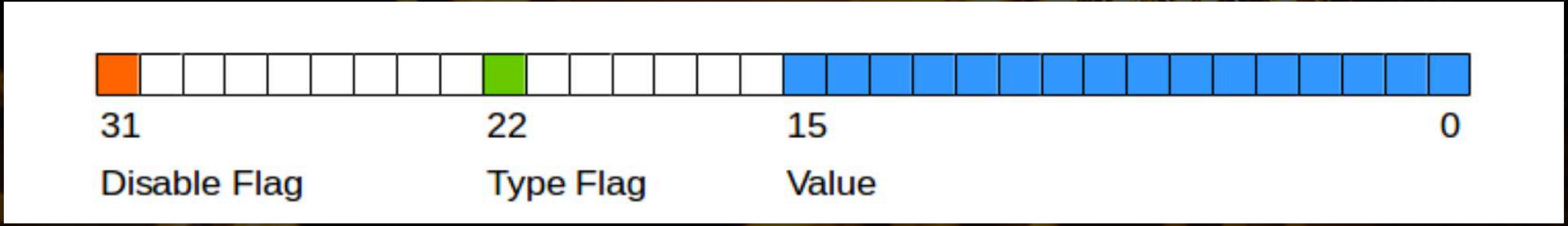
<= 0xFFFFFFFE — Locktime.

<= 0xFFFFFFFD — Replace By Fee (RBF).

<= 0xEFFFFFFF — Relative Locktime.

0x00000000 to 0x0000FFFF —> Block Lock

0x00400000 to 0x0040FFFF —> Time Lock




OPEN ASSETS
PROTOCOL

OPEN ASSETS

- Uses OP_RETURN and OP_CHECKMULTISIG
- Has memory overhead
- More flexible
- Can use Bit-Torrent for large data storage

Issuance Transaction Encoding

Bytes	Description	Comments	Stored in
2	Protocol Identifier	`0x4343` ASCII representation of the string CC ("Colored Coins")	OP_RETURN
1	Version Number	Currently `0x02`	OP_RETURN
1	**Issuance** OP_CODEs		OP_RETURN
20	SHA1 Torrent Hash	(*optional*), only when metadata is included	OP_RETURN or (1|**3**) Multisig
32	SHA256 of metadata	(*optional*), only when metadata is included Allows for torrent metadata verification	OP_RETURN or (1|**2**) or (1|**3**) Multisig
1-7	Amount of **issued units**	Encoded with the [Encoding Scheme](Number Encoding)	OP_RETURN
2-9 (per instruction)	[Transfer Instruction](Transfer Instructions)	Encoding the flow of assets from inputs to outputs	OP_RETURN
1	Issuance Flags	At the moment only 6 bits are used	OP_RETURN

Transfer Transaction Encoding

Bytes	Description	Comments	Stored in
2	Protocol Identifier	`0x4343` ASCII representation of the string CC ("Colored Coins")	OP_RETURN
1	Version Number	Currently `0x02`	OP_RETURN
1	**Transfer** OP_CODEs		OP_RETURN
20	SHA1 Torrent Hash	(*optional*), only when metadata is included	OP_RETURN or (1|**3**) Multisig
32	SHA256 of metadata	(*optional*), only when metadata is included Allows for torrent metadata verification	OP_RETURN or (1|**2**) or (1|**3**) Multisig
2-9 (per instruction)	[Transfer Instruction](Transfer Instructions)	Encoding the flow of assets from inputs to outputs	OP_RETURN

FUTURE WORK


THANK
YOU